

ADITI GUPTA

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SUMMARY

AI/ML undergraduate with research at TCS Performance Lab and FOSSEE, IIT Bombay. Skilled in machine learning, deep learning, generative AI, and computer vision, with ability to build end-to-end AI pipelines, optimize cloud systems, and deliver measurable results. Awarded research honorarium for contributions to deep learning and machine learning research in predictive modeling systems.

EDUCATION

Vellore Institute of Technology Bhopal University

B.Tech in CSE (Specialization in AI & ML) | CGPA: 8.87/10

Bhopal, India

Aug 2023 – Present

DAV Public School

Senior Secondary Education (Grade XII), CBSE – 93.2%

Gurugram, India

Apr 2022 – Mar 2023

Delhi Public School

Matriculation (Grade X), CBSE – 92.6%

Gurugram, India

Apr 2020 – Mar 2021

SKILLS

Programming Languages: Python, C++, Java, SQL

CS Fundamentals: Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLM

Frameworks & Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, PyTorch, Keras, MediaPipe, OpenCV

Tools & Platforms: Flask, Streamlit, AWS, Render, Git, GitHub, Jupyter Notebook, VS Code, DWSIM, Google Vertex AI, Docker

EXPERIENCE

TCS, Performance Research Lab | Research & Innovation Summer Intern | Mumbai

May 2026 – July 2026

Systems Optimization in Generative AI Based Applications

- Enhancing a cloud based Accelerator Recommendation system for LLMs that analyzes client hardware specifications and workload requirements to recommend optimal accelerator configurations.
- Optimising systems through compute estimation and CSP recommendation (AWS, GCP, Azure) for scalable AI deployment.

FOSSEE, IIT Bombay | Machine Learning Intern | Remote

Oct 2025 – Feb 2026

Crude Oil Characterization & Product Properties Estimation | [GitHub](#) | [Certificate](#) | [Published Work \(FOSSEE\)](#)

Python, XGBoost, Scikit-Learn, TensorFlow, Keras

- Built a data extraction and validation pipeline for 110+ global crude oil assay files with domain-specific checks across hydrocarbon composition, kinematic viscosity, and quality indices.
- Benchmarked 12+ ML algorithms, achieving R^2 up to 0.91 (XGBoost) for kinematic viscosity and 0.73 (SVR) for hydrocarbon composition; work published on the FOSSEE website; awarded honorarium by FOSSEE, IIT Bombay for contribution to the project.

PROJECTS

Real-Time Indian Sign Language Fingerspelling Translator | [GitHub](#)

Apr 2026

Python, Scikit-Learn (MLPClassifier), MediaPipe, OpenCV

- Built a real-time Indian Sign Language fingerspelling translator recognizing all 26 ISL alphabet gestures (A–Z) with hold-to-confirm word-building logic at 30+ FPS on standard consumer hardware.
- Engineered a 63-dimensional landmark-based feature pipeline using 21 MediaPipe hand points and trained a 3-layer MLPClassifier (256→128→64) on 5,277 samples from 7,800 ISLRTC images, achieving 96.69% test accuracy.

Binary Distillation Surrogate Model | [GitHub](#)

Sep 2025

Python, XGBoost, Random Forest, DWSIM, Scikit-Learn

- Developed ML surrogate models replacing expensive DWSIM simulations to predict distillate purity (xD) and reboiler duty (QR); XGBoost achieved R^2 of 0.95 and 0.93 respectively.
- Generated a 373-sample dataset via DWSIM simulations varying 6 operating parameters and benchmarked 6 regression models using RandomizedSearchCV; project secured selection for the FOSSEE IIT Bombay internship.

Student Performance Predictor | [GitHub](#) | [Live Demo](#)

Sep 2025

Python, CatBoost, XGBoost, Flask, Render

- Built an end-to-end ML pipeline and automated preprocessing with Scikit-Learn ColumnTransformer (StandardScaler + OneHotEncoder) across 1,000 records and 8 features.
- Evaluated 10+ regression algorithms with GridSearchCV; Ridge Regression achieved R^2 of 0.88. Deployed as a production Flask web app on Render with a responsive UI for real-time predictions.

CERTIFICATIONS

- Applied Machine Learning in Python** – University of Michigan
- Internet of Things Certification (Elite + Gold)** – NPTEL | Top 5% National
- Cloud Computing Certification (Elite)** – NPTEL

ACHIEVEMENTS

Awarded Research Honorarium by FOSSEE, IIT Bombay | Served as Head Girl and School Captain at Delhi Public School